2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$	$\sum y_i =$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.

$\hat{y}_i = 1.2 + 0.8x_i$

Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
–1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i\hat{e}_i = 0$

a. $\bar{x} = 1$ $\quad \bar{y} = 2$

b. The Ordinary Least Squares (OLS) Estimators

$$b_2 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \qquad b_1 = \bar{y} - b_2\bar{x}$$

$$= \frac{8}{10} = 0.8 \qquad = 2 - 0.8 \times 1 = 1.2$$

When x increases by one unit, y is expected to increase by 0.8 units on average.

c. $\Sigma X_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$

$\Sigma X_i Y_i = 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0 = 18$

Show ① $\Sigma X_i^2 - N\bar{X}^2 = 15 - 5 \times 1^2 = 10$ equal to $\Sigma (X_i - \bar{X})^2$

② $\Sigma X_i Y_i - N\bar{X}\bar{Y} = 18 - 5 \times 1 \times 2 = 8$ equal to $\Sigma (X_i - \bar{X})(Y_i - \bar{Y})$

d.

- $S_y^2 = \dfrac{\Sigma (Y_i - \bar{Y})^2}{N-1} = \dfrac{2^2 + 0^2 + 1^2 + (-1)^2 + (-2)^2}{5-1} = \dfrac{10}{4} = 2{,}5$

- $S_x^2 = \dfrac{\Sigma (X_i - \bar{X})^2}{N-1} = \dfrac{10}{5-1} = 2{,}5$

- $S_{xy} = \dfrac{\Sigma (X_i - \bar{X})(Y_i - \bar{Y})}{N-1} = \dfrac{8}{5-1} = 2$

- $r_{xy} = \dfrac{S_{xy}}{S_x S_y} = \dfrac{2}{\sqrt{2{,}5 \times 2{,}5}} = 0{,}8$

- $CV_x = 100 \cdot \dfrac{S_x}{\bar{X}} = 100 \cdot \dfrac{\sqrt{2{,}5}}{1} \approx 158{,}1\%$

- median $= -1, 0, 1, 2, 3$

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

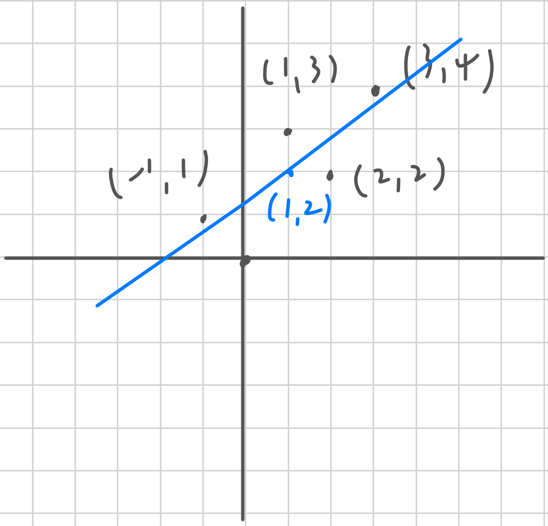
g. Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\text{var}}(b_2 | \mathbf{x})$ and $\text{se}(b_2)$.

e. f



$\hat{y} = 1.2 + 0.8x$

$(\bar{x}, \bar{y}) = (1, 2)$

g. Show $\bar{y} = b_1 + b_2 \bar{x} = 1.2 + 0.8 \times 1 = 2$

$\bar{y} = 2$ (By OLS $b_1 = \bar{y} - b_2 \bar{x}$) QED

h. $\bar{\hat{y}} = \frac{\Sigma \hat{y}}{N} = \frac{10}{5} = 2$ $\bar{y} = \bar{\hat{y}}$ QED

i. $\hat{\sigma}^2 = \frac{\Sigma \hat{e}_i^2}{N-K} = \frac{3.6}{5-2} = 1.2$

j. $\widehat{\text{var}}(b_2 | x) = \frac{\hat{\sigma}^2}{\Sigma (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$\text{se}(b_2) = \sqrt{\widehat{\text{var}}(b_2 | x)} = \sqrt{0.12} \approx 0.3464$

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

a. $\widehat{TOTALSCORE}_i = 916.442 + 12.175 \times SMALL_i$

Intercept $\beta_1 = 916.442 \Rightarrow$ when $SMALL = 0$, the average combined aptitude scores of students in the regular class is 916.442

Slope $\beta_2 = 12.175 \Rightarrow$ when $SMALL = 1$, the average combined aptitude scores of students in the small class is 12.175 points higher than that of students in the regular class.

The p-value 0.0236 for the smaller class, is less than 0.05

This means that the difference of 12.175 is statistically significant

Conclusion: Reducing class size has a significant positive effect on students' overall learing scores.

b.

$READSCORE_i = 432.665 + 6.924\ SMALL_i$

In Reading, small class sizes improved reading scores by an average of 6.924 points. and the p-value is 0.00151 (less than 0.05) is statistically highly significant

$MATHSCORE_i = 483.777 + 5.251\ SMALL_i$

In Math. small classes increased math scores by an average of 5.251 points but the p-value is 0.143 (greater than 0.05) not statistically significant.

Conclusion: The learning benefits of smaller classes vary across different subjects. Smaller classes improve reading score, but math scores do not.

c. $TOTALSCORE_i = 916.442 + 4.306\ AIDE_i$

Intercept 916.442 ⇒ when $AIDE_i = 0$. the average combined aptitude scores of students in the regular class without teaching assistants is 916.442

Slope 4.306 ⇒ when $AIDE_i = 1$. the average combined aptitude scores of students in the regular class with teaching assistants is 4.306 points higher than without assistants.

AIDE

Conclusion: The p-value for teaching assistants variable is 0.389. which is significantly greater than the 0.05 significance level. This indicates maybe is random error. ⇒ not statistically significant

Adding a full-time teaching assistant to a regular class does not significantly improve students' overall learning performance

d.

$READSCORE_i = 432.665 + 2.871\ AIDE_i$

Adding a teaching assistant increases the reading score by 2.871 points.

With p-value of 0.161 (greater than 0.5) ⇒ not significant

$MATHSCORE_i = 483.777 + 1.435\ AIDE_i$

Adding a teaching assistant increases the math score by 1.435 points.

With p-value of 0.664 (greater than 0.5) ⇒ not significant

Conclusion: Unlike small classes, which are effective in reading subjects, adding teaching assistants does not improve in reading or math scores.

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between \$1000 per month to \$20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

a. Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?

b. What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

c. Construct a histogram of ln(*FOODAWAY*) and its summary statistics. Explain why *FOODAWAY* and ln(*FOODAWAY*) have different numbers of observations.

d. Estimate the linear regression $\ln(FOODAWAY) = \beta_1 + \beta_2 INCOME + e$. Interpret the estimated slope.

e. Plot ln(*FOODAWAY*) against *INCOME*, and include the fitted line from part (d).

f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

a.

Min	1st Q	Median	Mean	3rd Q	Max
0	12.04	32.55	49.27	67.5	1179.0

25% quantile = 12.04   75% quantile = 67.5025

b. Household with higher education levels have higher food-away from-home expenditures

Advanced Degree   Mean = 73.15   Median = 48.15

College Degree   Mean = 48.6   Median = 36.11

No Degree   Mean = 39.01   Median = 26.02

c.

Cuz the original FOODAWAY data includes observations with a value of 0.
Mathematically, the natural log (ln(0)) is undefined.
These values cannot be calculated when performing statistical analysis or plotting involve log transformations. Therefore, it must be discarded, resulting in fewer

vaild observations for ln(FOODAWAY) than for the original variables.

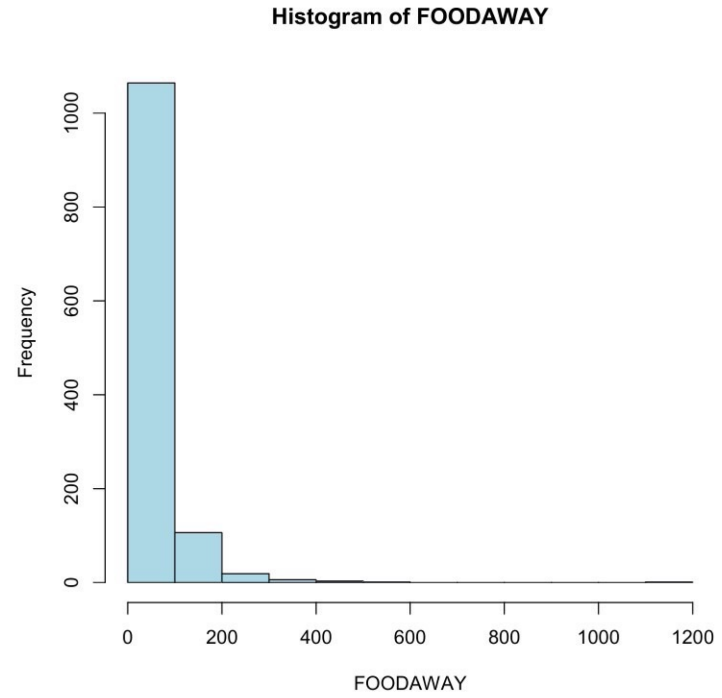
d. $\widehat{\ln(FOODAWAY)} = 3.1293 + 0.0069017 \times INCOME$

slope: 0.0069017

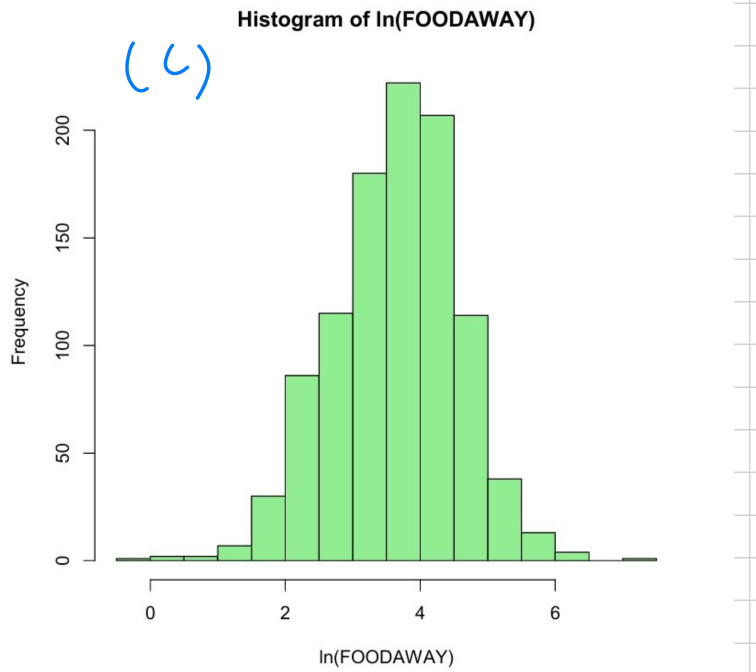
In log-linear model. the slope represents the percentage change expected in the independent variable when the independent variable change by 1 unit. When household mothly income (INCOME) increase by 1 unit ($100) monthly spending on eating out per person is expected to increase by 0.69% (p-value = 2e^(-16) ⇒ statistically significant)

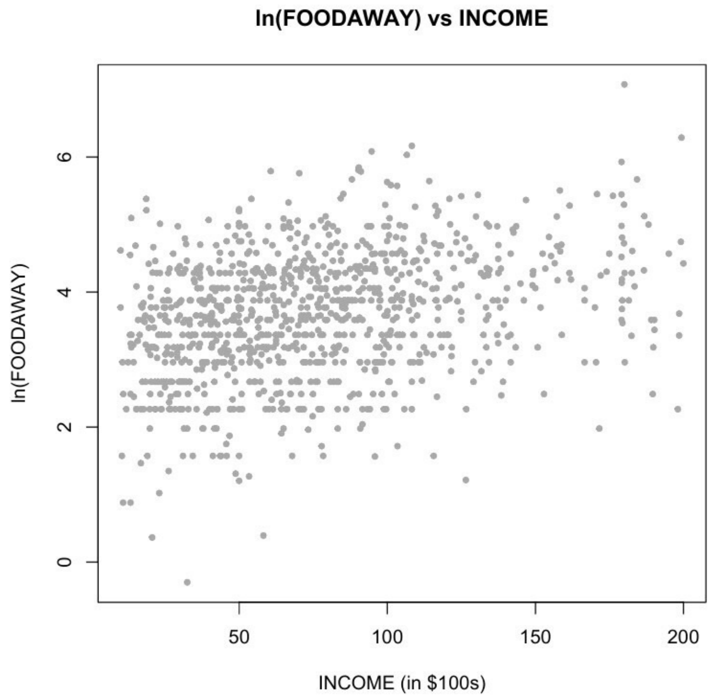
f.

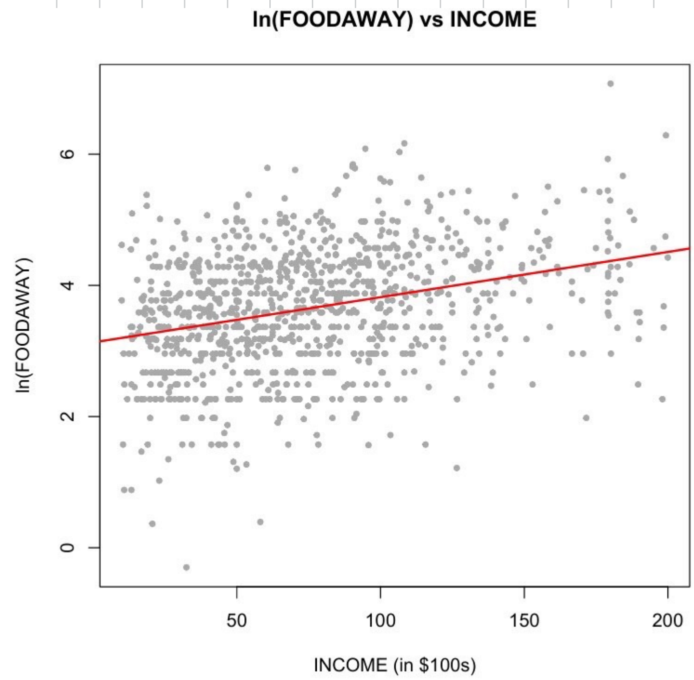
in this type of cross-sectional consumption data. the residuals are not completely random. Notice that as INCOME increase, the vertical spread of the residuals also gets wide (funnel shape) ⇒ Heteroskedasticity


Histogram of FOODAWAY
Frequency
FOODAWAY

(c)


Histogram of ln(FOODAWAY)
Frequency
ln(FOODAWAY)


ln(FOODAWAY) vs INCOME
ln(FOODAWAY)
INCOME (in $100s)


ln(FOODAWAY) vs INCOME
ln(FOODAWAY)
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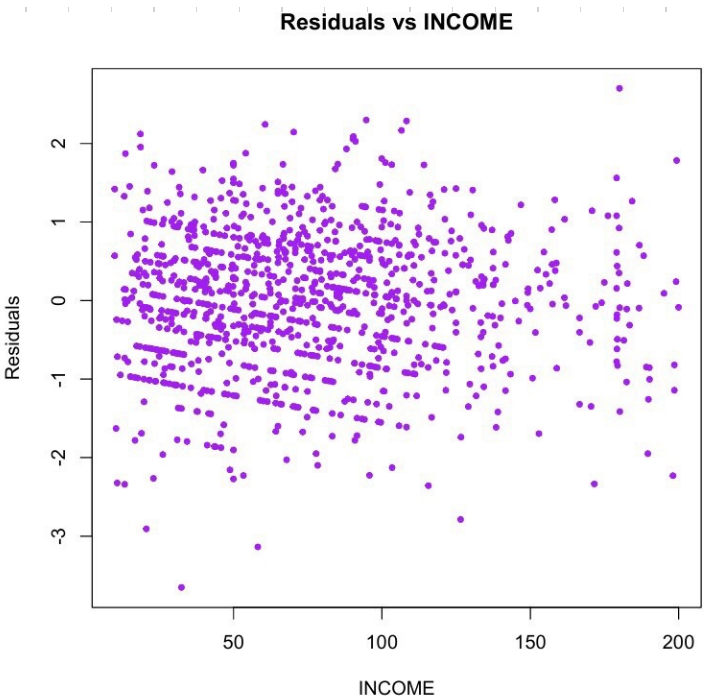
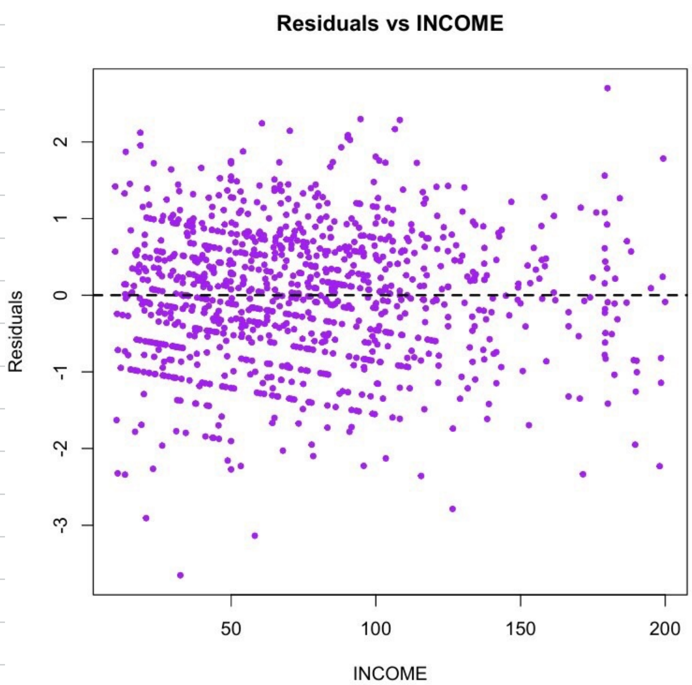

Residuals vs INCOME
Residuals
INCOME


Residuals vs INCOME
Residuals
INCOME